PROBLEM7:

PROBLEM STATEMENT:

Retail Billing with Multiple Discount Rules: Process multiple products, quantities, category specific taxes, threshold discounts, and maximum concession limits to generate an itemized invoice.

OBJECTIVE:

To develop a Python program that generates an itemized retail bill by processing multiple products and quantities, applying category-specific taxes, threshold-based discounts, and maximum concession limits, and calculating the final amount payable by the customer.

PROGRAM:

```
# Retail Billing with Multiple Discount Rulesn = int(input("Enter number of products: "))subtotal = 0total_tax = 0print("\n----- ITEMIZED INVOICE -----")for i in range(n): print("\nProduct", i + 1)  name = input("Enter product name: ")  category = input("Enter category (food/electronics/clothing): ").lower()  price = float(input("Enter price per item: "))  quantity = int(input("Enter quantity: "))  amount = price * quantity  # Category-specific tax  if category == "food":  tax_rate = 5  elif category == "electronics":  tax_rate = 18  elif category == "clothing":  tax_rate = 12  else:  tax_rate = 10  tax = amount * tax_rate / 100  subtotal += amount  total_tax += tax  print(f"{name}: {quantity} x Rs.{price:.2f}")  print(f"Amount: Rs.{amount:.2f}")  print(f"Tax ({tax_rate}%): Rs.{tax:.2f}")# Threshold-based discountif subtotal >= 5000:  discount_rate = 15elif subtotal >= 2000:  discount_rate = 10elif subtotal >= 1000:  discount_rate = 5else:
```

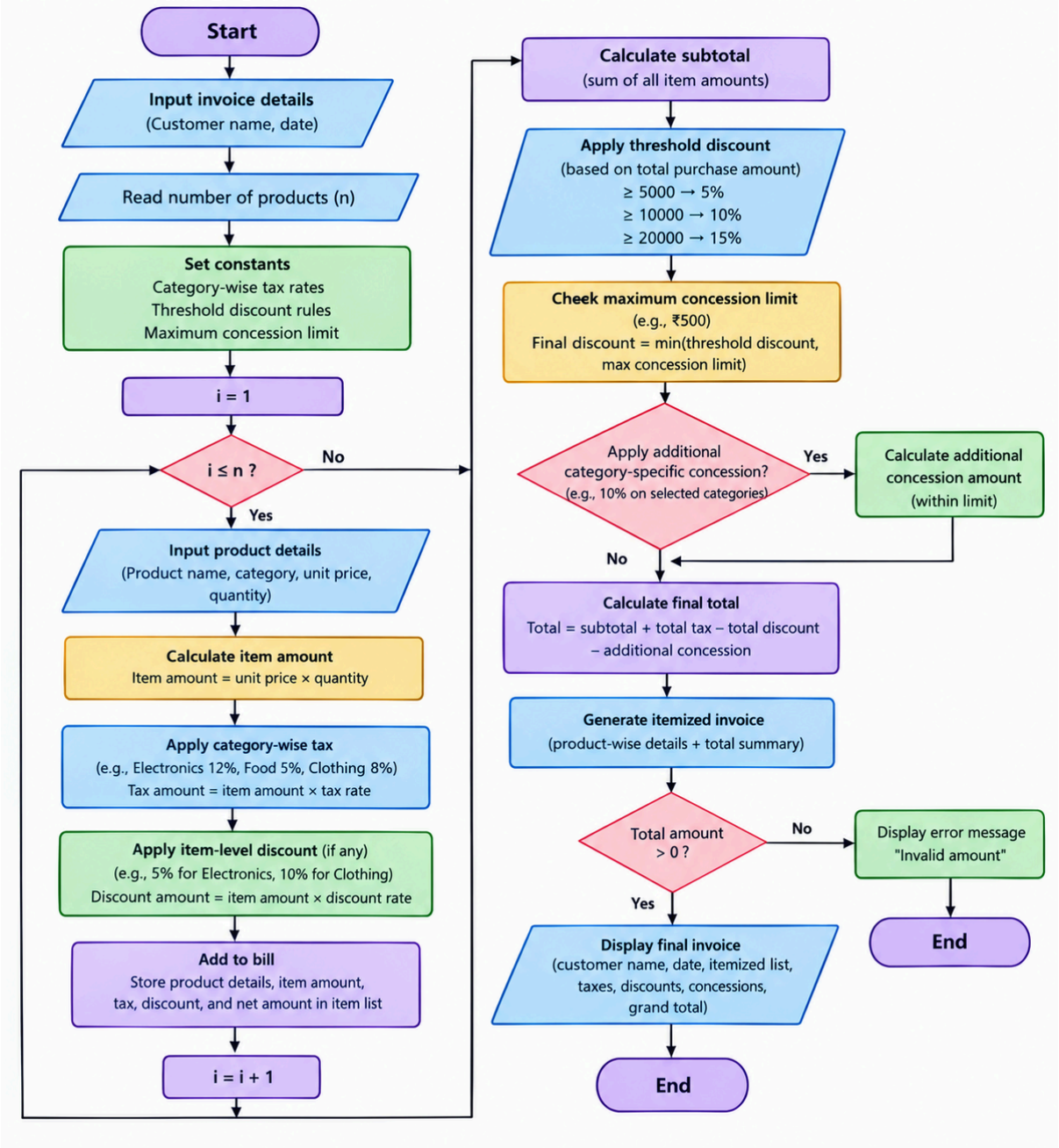
ALGORITHM:

1. Start the program.
2. Read the number of products.
3. Initialize subtotal and total tax to zero.
4. For each product:
    - Read the product name, category, price, and quantity.
    - Calculate the item amount as price × quantity.
    - Determine the tax rate based on the category.
    - Calculate the tax and update the subtotal and total tax.
    - Display the item details.
5. Apply the discount based on the subtotal:
    - 15% if subtotal is ₹5,000 or more.
    - 10% if subtotal is ₹2,000 or more.

- 5% if subtotal is ₹1,000 or more.
- Otherwise, no discount.

6. Limit the discount to a maximum of ₹1,000.
7. Calculate the final bill as subtotal + total tax − discount.
8. Display the itemized invoice and final amount.
9. Stop.

FLOWCHART:



PSEUDOCODE:

```
START

INPUT number_of_products
subtotal = 0
total_tax = 0

FOR each product:
    INPUT name, category, price, quantity
    amount = price * quantity

    IF category = "food":
        tax_rate = 5
    ELSE IF category = "electronics":
        tax_rate = 18
    ELSE IF category = "clothing":
        tax_rate = 12
    ELSE:
        tax_rate = 10

    tax = amount * tax_rate / 100
    subtotal = subtotal + amount
    total_tax = total_tax + tax

    DISPLAY name, amount, tax

IF subtotal >= 5000:
    discount_rate = 15
ELSE IF subtotal >= 2000:
    discount_rate = 10
ELSE IF subtotal >= 1000:
    discount_rate = 5
ELSE:
    discount_rate = 0

discount = subtotal * discount_rate / 100

IF discount > 1000:
    discount = 1000

final_amount = subtotal + total_tax - discount

DISPLAY subtotal
DISPLAY total_tax
DISPLAY discount
DISPLAY final_amount

STOP
```

OUTPUT:

RETAIL STORE INVOICE

Sample bill for 3 products

Product	Amount	Tax
Rice (Food)	₹1,500.00	₹75.00
Headphones (Electronics)	₹3,000.00	₹540.00
Shirt (Clothing)	₹1,000.00	₹120.00

Subtotal

₹5,500.00

Total Tax

₹735.00

Discount (15%)

−₹825.00

Final Amount Payable

₹5,410.00

The sample discount is ₹825 because 15% of ₹5,500 is ₹825, which is below the maximum concession of ₹1,000.